

Ejercicio 9

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-- Ejercicio 14
foldNat :: (Integer -> b -> b)
         -> b
         -> Integer
         -> b
foldNat fRec fBase x = case x of
  0 -> fBase
  n -> fRec n (foldNat fRec fBase (n-1))

potencia :: Integer -> Integer -> Integer
potencia n = foldNat (\i rec -> n * rec) 1
```

Defino $P(n)$: potencia x $n = x^n$

Caso base $n=0$

$P(0)$: potencia x $0 = x^0$

$$1 = 1 \quad \checkmark$$

Paso Inductivo

Busco probar que $\forall n :: \text{Nat} \quad P(n) \Rightarrow P(n+1)$

$\{HI\} P(n)$: potencia x $n = x^n$

$$\text{potencia } x(n+1) = x^{n+1}$$

$$\stackrel{\{PO\}}{=} \text{foldNat } (\lambda i \text{ rec} \rightarrow x * \text{rec}) 1 (n+1)$$

$$\stackrel{\{FN\}}{=} (\lambda i \text{ rec} \rightarrow x * \text{rec}) (n+1) (\text{foldNat fRec 1 } n)$$

$$\stackrel{\{HI\}}{=} (\lambda i \text{ rec} \rightarrow x * \text{rec}) (n+1) x^n$$

$$\stackrel{\{P\}}{=} x^n \cdot x = x^{n+1} \quad \square$$